

NAKSHA

An augmented reality service that plans a home's wiring, and sells the wire

Track 4: Digital Solutions for Consumer Engagement

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Synopsis

A family building a house does not know where anything should go. Not the sockets, not the switchboards, not the water heater point. The architect's drawing, if there is one, says almost nothing about services, so the decisions get made on site by whoever is standing there. Runs are improvised, material is mis-ordered, and nothing is written down. Around 42% of building fires in India are attributed to electrical faults [1], and with no record of what was laid, any later renovation is guesswork.

NAKSHA is an app from V-Guard that closes that gap. The owner answers a short set of questions, scans each room with the phone's LiDAR sensor, and receives a complete electrical design: where every light, fan, socket and switchboard goes, how the points group into circuits, what cable and breaker each circuit needs, and how many metres of wire the house will consume. The same design is then projected onto the real walls in augmented reality, so it can be seen at full size before anything is chased or plastered.

The design itself is not produced by a language model. A rule engine does it, so every decision can be traced. The model's only job is the conversation.

For the owner this replaces total dependence on one electrician with a drawing, a quantity and a permanent record. For V-Guard it is a sales channel rather than a product: it can be given away free because the revenue is the wire, the water heater and the chimney it specifies. A working prototype exists. It designs a real surveyed room, and across three house types it shows that routing conduit as a shared tree instead of one run per point saves a mean of 45.8% of cable length, with a further 7.7% from siting the board properly.

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1. Story time

A few years ago my family used Berger's interior painting service. It was expensive. It was not available in most pincodes. We used it in one room of the house and nowhere else.

My father still curses the price of that paint. Years later, he remembers the brand exactly.

That is the whole insight. The service never scaled. It did not need to. What it did was put a premium brand physically inside our house for two days, and that memory outlasted every advertisement Berger ran in the same period. Asian Paints had already proved the mechanism with Beautiful Homes, and Berger followed it with Express Painting [2]. Neither company was trying to become a painting contractor. They were buying a position.

V-Guard is in a similar place, with a harder problem. It sells wires, water heaters, chimneys, RO purifiers, pumps and gas stoves. It is trusted. It is not seen as the brand you consult before the walls close. On the Forbes India and TRA Research list of most respected consumer technology brands, V-Guard sits behind Havells, Bajaj Electricals, Syska and Cona [3]. That is a brand with permission it is not using.

NAKSHA is the same move as Beautiful Homes, except it costs software money rather than field money, and it is aimed at the one moment when a home owner has to decide which brand goes into the wall.

2. The problem

This is sharpest outside the metros, which is also where the growth is. Land prices in tier 2 and tier 3 cities are expected to rise 25% to 100% over the next two to four years as metro prices peak [4], and those are markets where houses are built by the owner rather than by a developer.

- There is often no architect. Where there is a drawing, it shows walls and rooms, not services.
- The electrician plans the installation, executes it, and is the only person who knows what was done.
- Nothing is recorded. Once the plaster is on, the information is gone.
- So the new owner has to think ten times before drilling a single hole in his own wall.

The consequences are not only inconvenience. Roughly 42% of building fires in India are attributed to electrical faults [1], and a NITI Aayog survey found 28% of households dissatisfied with the quality of their electricity supply as recently as 2023 [5]. Sockets end up where the furniture is not. Air conditioners get added later onto circuits that were never sized for them.

For V-Guard the same moment is a commercial one. Every one of those decisions settles which brand of wire, water heater, chimney and stove ends up in the house. V-Guard sells all of them and is present for none of it.

3. What does the idea do?

The owner opens the app. It asks a short series of questions: name, sanctioned load from the electricity bill, how many bedrooms, how many people, how many air conditioners and in which rooms, which bathrooms need water heating, what the kitchen will have. Roughly eight questions, each one written from the answers already given.

Then the owner walks each room holding the phone. The LiDAR sensor measures the walls, the doors, the windows and the furniture. Nothing is typed in.

From the room and the answers the app produces four things:

- A drawing. Every light, fan, socket, switchboard and appliance point positioned, with the conduit routed between them and the distribution board sited.
- A circuit schedule. Which points share a circuit, the breaker rating, the cable size, the connected load and the voltage drop on each.
- A quantity. Metres of cable by size, metres of conduit, counts of boxes and switchgear, and what it costs.
- An augmented reality view. The same design projected onto the actual walls and ceiling at full size, so it can be walked through before anything is cut.

As each point is confirmed on site, it is ticked off. What remains at the end is an as-built record: a permanent map of where the cables actually went, which is the thing that does not exist today.

4. How it works

There are three parts, and the division between them is deliberate.

4.1 The conversation

A language model asks the questions. It runs on a server, holds no engineering knowledge, and produces one thing: a profile of the household. Its output is a structured record of what the family intends to own, not a wiring decision.

This boundary matters more than anything else in the design. A model that hallucinates a cable size is a fire. So the model is not allowed near the engineering, and if the model is unreachable a fixed set of questions runs instead and the app behaves identically.

4.2 The design engine

Everything that could hurt somebody happens here, in explicit rules.

Lighting counts come from the lumen method against the measured floor area and an illuminance target for the room type. Socket counts follow spacing rules. Points are grouped into lighting, power and dedicated circuits. The distribution board is then placed by solving for the position that minimises total cable, and conduit is routed between the points as a rectilinear Steiner minimal tree, which is the same problem printed circuit board routers solve [6].

Sizing is done in a specific order: the breaker is chosen first from the design current, then a conductor whose ampacity covers that breaker. The reverse order is a protection violation, because a breaker rated above the cable it protects will not trip before the cable overheats. Voltage drop is then checked and the conductor increased if it fails. Maximum demand is calculated with diversity factors and compared against the sanctioned load, so a house with three air conditioners on a 5 kW sanction is told so while the walls are still open.

4.3 The augmented reality view

The design is authored on a floor plane in metres. To place it in the room, the app needs to know where the room is. The user points at two corners of the floor and taps each, which fixes position and rotation exactly, and the ceiling height comes from the scan. Registration is manual and visible on purpose. Automatic alignment drifts on a live building site, and an overlay that is confidently in the wrong place is worse than one the user positioned himself. The authoritative output is always the dimensioned drawing, never the phone screen.

5. What does this mean for the user?

- He is no longer completely dependent on one electrician or contractor. He has a drawing, and a second opinion is now possible.
- He has a 3D and augmented reality map of his own wiring, kept for the life of the building. Drilling a hole stops being a gamble.
- He knows how much wire and how much switchgear the house will consume, so he buys the right quantity and overcharging becomes visible.
- He sees a brand that turned up before the wall was closed rather than after it failed. That is what a premium brand looks like from inside a house.

The electrician is not displaced by this, and the design does not assume he will be. He remains legally responsible for the installation. What he gets is the load calculation done, the sizes decided, the material list totalled, and a defensible answer when the owner asks why it costs what it costs.

6. What does V-Guard get from all this?

Four things, in order of how much they are worth.

- V-Guard stops being only a trusted brand and becomes a technology first one. That repositioning is very hard to buy with advertising and reasonably cheap to buy with a working app.
- Every design is an order. The output is a material schedule in metres and units that can be mapped to V-Guard product codes. Specifying material is a stronger commercial position than advertising to whoever walks into a shop.
- The decision is captured years earlier. Services are chosen during construction, before the owner has formed any brand opinion, and they are never re-shopped. Nobody rewires a finished house.
- If the app reaches ordinary customers and local electricians, V-Guard gets first-hand market information: what is actually being installed, in which cities, at what load, in what quantity. Nobody in this category has that today.

The margin case is the reason this is worth doing at all. In FY26 V-Guard's revenue grew about 7% to Rs 5,966 crore while profit after tax fell 1.7% to Rs 308 crore, with an EBITDA margin of 8.8% [7]. The electricals segment, which contains wires, is more than 40% of revenue [8]. Meanwhile Polycab holds roughly a quarter of a cables and wires market worth about Rs 1 lakh crore [9][10]. V-Guard cannot win that market on price against a competitor several times its size in the category. Buying share with discounts is what compresses the margin further. Being specified on a drawing before anyone reaches the counter is the alternative.

There is also a tailwind. The organised share of the cables and wires industry rose from about 67% in FY22 to about 80% in FY26 [10]. NAKSHA accelerates exactly that shift, because a documented design forces certified branded material. You cannot write a conductor size on a drawing and then buy loose unbranded wire against it.

And the channel already exists. V-Guard runs Rishta, a loyalty programme for electricians and plumbers with QR scanning and instant payouts [11]. Every major wire brand runs something similar. All of them are points and rewards schemes. None of them puts a tool in the electrician's hand that does any actual work. NAKSHA can be distributed inside a channel V-Guard owns.

7. Does it actually work?

A prototype exists: an iOS application and a design engine. It has been run on a real room.

The room is 15 feet by 12 feet with a 10 foot ceiling, surveyed and cross checked against a mesh scan which measured the footprint to within 1.4%. It contains two lights, a ceiling fan, three switchboards, three sockets, an air conditioner and an existing MCB box. From those eleven points the engine produced three circuits: a 6 A lighting circuit on 1.5 sq mm, a 16 A power circuit on 2.5 sq mm, and a dedicated 25 A circuit for the air conditioner on 4 sq mm. It routed 41 metres of conduit, reported a maximum demand of 1,959 W against a 4,500 W sanction, and priced the material at Rs 12,081. No code violations were raised.

The 25 A circuit is worth a sentence, because it shows the ordering rule doing its job. A one ton air conditioner would normally be given a 16 A breaker. The isolator physically fitted in that room is 25 A, so the engine took the larger device and raised the conductor to 4 sq mm, whose ampacity is exactly 25 A. A 2.5 sq mm cable under a 25 A breaker would have been a protection violation.

The quantitative result is the routing study. Three house types were designed twice, once with a separate run to every point and once with conduit shared as a Steiner tree:

House type	Radial	Steiner	Topology saving	Board siting
1 BHK, 48 sq m	201.3 m	114.8 m	37.5%	8.7%
2 BHK, 84 sq m	504.0 m	253.6 m	44.9%	8.6%
3 BHK, 130 sq m	822.8 m	349.3 m	54.9%	5.8%
Mean			45.8%	7.7%

Exhibit 1. Conduit length by routing method, measured on three house types. Radial is one run per point. Topology saving is the reduction from sharing conduit as a Steiner tree. Board siting is the further reduction from solving for the board position, holding topology constant.

The first saving is a mean of 45.8% of conduit length from sharing the topology, and the second is a further 7.7% from letting the engine site the distribution board rather than defaulting to the entry door. The second figure is the more honest of the two. Experienced electricians already loop lighting points in and out, which recovers part of the topology saving, so 45.8% should be read as an upper bound against the worst case. The board siting figure holds the topology constant and is a clean result.

The cost of delivery is software cost. There is no field organisation, no equipment to maintain, and no trained staff to place in every pincode, which is precisely why this approach was chosen over anything involving hardware on site.

8. What it does not do yet

Stating this plainly is more useful than claiming otherwise.

- The scan captures one room at a time and separate captures do not share a coordinate frame. Room shapes and areas are measured; the arrangement between rooms is approximated. The schedule and the quantities are unaffected because they depend on area and room type.
- Augmented reality accuracy on a live site is unproven. The promise is dimensioned guidance and reference marks, not millimetre placement.
- Only the electrical layer is built. Water, gas and duct routing use the same geometry and the same router, but they are not implemented.
- Liability needs legal structuring. The tool must be a decision aid a licensed contractor signs off, not the designer of record.
- Adoption, not technology, is the real risk. If the app does not visibly save an electrician time on load calculation and estimating from the first use, it will be ignored by the people who decide what gets bought.

9. Why this is worth doing

The components exist and the proposal is stronger for saying so. Automated electrical takeoff is a mature category, with tools such as drawer.ai, Kreo and Trimble Accubid serving contractors who bid large projects [12]. Robotic site layout is solved, with Dusty Robotics having printed coordinated models onto hundreds of millions of square feet of slab [13].

All of it serves Western commercial contractors. The contribution here is positional. The user is the home owner rather than an estimator, so the design is created by walking around rather than uploaded as a finished drawing. Indian practice and Indian codes are the rule set. The as-built record is produced, which no takeoff tool does, because its users leave the site and never return. And the whole thing is monetised by a company that sells every product it touches.

The construction sector is the single largest consumer of cable in India at about 32% of demand [14], and residential construction is growing at about 6.6% a year [15]. That is the pipeline. NAKSHA puts V-Guard at the front of it, in the hand of the person deciding, at the one moment the decision is still open.

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